A Constant Vector Beats Your Digital Twin: On the Evaluation of Cluster-Level LLM Personality Simulation

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Abstract

Large language models are increasingly prompted to act as "digital twins" of human personality groups, and such systems are routinely evaluated by how closely the model's questionnaire answers match those of real respondents. We show that this evaluation is broken. For the answer-similarity metric used throughout this literature, we prove that a simulator which reproduces the target population's response distribution *exactly* is strictly dominated by a constant vector, and that the gap is governed by the difference between the Gini mean difference and the mean absolute deviation about the median. On $410{,}168$ IPIP-NEO-120 respondents this structural penalty is 7.75 points. Empirically, across five LLMs and four personality clusters, a per-item cluster mean scores 0.796 while the best evolutionary-optimised persona reaches 0.739, losing in 20 of 20 cells; a *real human drawn from the target cluster* scores 0.721 and also loses to the constant in 20 of 20 cells. We further find that the released benchmark scores raw model self-reports against reverse-recoded human answers on 55 of 120 items; we correct this and report every result under both orientations. A metric that ranks a constant above a genuine group member cannot be measuring personality simulation. We propose a replacement suite — a classifier two-sample test, per-item distributional fidelity, and a variance ratio — each reported against an explicit human ceiling and constant floor, and show it inverts the ranking correctly. Under it, current personas are near-perfectly separable from humans (C2ST-AUC 0.983 vs. a human ceiling of 0.564) and produce 43% of human response variance. We release code and per-cell results.

1 Introduction

Simulating human respondents with large language models has become a common instrument in computational social science and in LLM-based user research. A widely used recipe conditions a model on a natural-language personality description and validates it by administering a psychometric inventory: the model answers the same Likert items a human answered, and the system is scored by the agreement between the two answer vectors.

This paper argues that the scoring step invalidates the conclusion. Our starting point is a concrete artefact. In a cluster-level digital-twin system built on IPIP-NEO-120 (Johnson, 2014), evolutionary prompt optimisation raises answer-level agreement while simultaneously *degrading* the recovered Big Five profile — mean absolute error, trait similarity and facet similarity all move the wrong way. That pattern is usually reported as an open puzzle. We show it is a predictable consequence of the metric.

Our contributions are:

1. **A proposition** (§3): under answer-similarity, a simulator that faithfully samples the target population's response distribution is *dominated* by the best constant predictor — strictly so on every IPIP-NEO item we measure. The penalty is the excess of the Gini mean difference over the mean absolute deviation about the median, divided by the scale range.
2. **An empirical audit** (§4): across five LLMs × four clusters, on the original system's own train/test splits, trivial constant baselines beat every model in every cell, and so does the global mean, which ignores the personality clusters entirely.
3. **A diagnosis of the human ceiling** (§8): a real respondent from the target cluster — the ideal a cluster-level twin can aspire to — also loses to the constant in every cell.
4. **A replacement suite** (§9): C2ST-AUC, distributional fidelity and variance ratio, each anchored by a human ceiling and a constant

floor. It ranks humans above constants, as any valid measure must.

5. **A cluster-validity audit** (§12): the k=4 partition these systems condition on is not supported by any internal criterion.

2 Setting

Task. A population of respondents is partitioned into K personality clusters by k-means over the 30 IPIP-NEO facet scores. For each cluster a natural-language description of the centroid is used to condition an LLM, which then answers the 120 IPIP-NEO items on a 1–5 Likert scale. Prompt wording is optimised by an evolutionary search whose fitness is answer agreement with held-out human members of the cluster.

The metric. A simulated answer vector x is scored against a human answer vector y by

$$S(x, y) \;=\; \frac{1}{J}\sum_{j=1}^{J}\Big(1 - \tfrac{|x_j - y_j|}{4}\Big), \qquad (1)$$

with J=120 and the divisor 4 the scale range. Cell scores average S over the held-out respondents. Equation (1), or a monotone transform of it, is the objective the search optimises and the number reported.

3 A constant is optimal under S

Fix a cluster and an item j, and let P_j be the cluster's response distribution over $\{1, \ldots, 5\}$. Write $Y \sim P_j$ for a randomly drawn human respondent.

Proposition 1. *Let m_j be a median of P_j, let $\mathrm{MAD}_j = \mathbb{E}_Y|m_j - Y|$ and let $\mathrm{GMD}_j = \mathbb{E}_{X,Y}|X - Y|$ for $X \sim P_j$ independent of Y (the Gini mean difference). Then*

- *(a) the best constant predictor attains expected score $1 - \mathrm{MAD}_j/4$;*
- *(b) a* faithful *simulator, i.e. one whose output is distributed as P_j, attains expected score $1 - \mathrm{GMD}_j/4$;*
- *(c) $\mathrm{GMD}_j \geq \mathrm{MAD}_j$, with equality iff X lies almost surely in the median set of P_j.*

Hence the constant is at least as good as the faithful simulator, and strictly better unless P_j places all its mass on medians.

Proof. (a) and (b) are immediate from (1). For (c),

$$\begin{aligned}\mathrm{GMD}_j &= \mathbb{E}_X\big[\mathbb{E}_Y|X - Y|\big]\\ &\geq \mathbb{E}_X\big[\min_c \mathbb{E}_Y|c - Y|\big]\\ &= \mathbb{E}_Y|m_j - Y| \;=\; \mathrm{MAD}_j,\end{aligned}$$

since the median minimises expected absolute deviation. Equality requires X to attain that minimum almost surely, i.e. to lie in the median set of P_j. This is weaker than degeneracy: a balanced two-point law $P(Y{=}1) = P(Y{=}5) = \frac{1}{2}$ has GMD = MAD = 2, so the bound is tight there despite large variance. On the IPIP items it is strict throughout (Table 2). □

The consequence is not a technicality. Under S, the ideal behaviour we actually want — reproducing the population's response distribution — is penalised relative to emitting one fixed vector. Optimising S therefore pushes a simulator toward degenerate central-tendency responding, destroying exactly the answer variance that carries a personality profile. This is the mechanism behind the "answer similarity up, profile fidelity down" pattern reported as a puzzle in prior work.

3.1 The metric is CRPS with its dispersion term deleted

Proposition 1 is not an accident of Likert data. The continuous ranked probability score for a predictive distribution F and outcome y is

$$\mathrm{CRPS}(F, y) = \mathbb{E}_F|X - y| - \tfrac{1}{2}\,\mathbb{E}_F|X - X'|,$$

and it is *strictly proper*: uniquely optimised by $F = P$. Equation (1) is its first term alone. Deleting the second is exactly what makes a constant optimal.

This yields the natural repair. Define

$$S_\lambda \;=\; 1 - \big[\,\mathbb{E}|X - y| - \lambda\,\mathbb{E}|X - X'|\,\big]/4. \quad (2)$$

λ=0 is the audited metric — improper, constant-optimal. $\lambda=1/2$ is CRPS — strictly proper, truth-optimal. $\lambda>1/2$ is improper in the opposite direction. Crucially $\lambda=1/2$ is **not** a tunable hyperparameter but the unique proper choice, so the repair cannot be dismissed as a metric tuned to produce a preferred conclusion.

Sweeping λ over the same 20 cells (Table 1) shows the repair works and what it costs the simulator. At $\lambda=1/2$ the human ceiling (0.859) rises decisively above the constant floor (0.796): the

λ	Model	Constant	Human
0 (audited)	0.660	**0.796**	0.721
0.25	0.678	**0.796**	0.775
0.5 (CRPS, proper)	0.697	0.796	**0.859**
0.75	0.720	0.796	**0.929**
1.0	0.744	0.796	**0.998**

Table 1: The S_λ family of (2), averaged over 20 cells. The audited metric sits at λ=0, where a constant wins. At the unique proper coefficient λ=1/2 a real cluster member wins, as it must. The simulator loses under both.

Clu.	n	MAD	GMD	S_{const}	S_{faith}
0	107,174	0.804	1.107	0.7990	0.7232
1	108,880	0.723	1.008	0.8192	0.7480
2	108,867	0.809	1.128	0.7978	0.7181
3	85,247	0.871	1.204	0.7822	0.6990
Mean		0.802	1.112	**0.7996**	0.7221

Table 2: Proposition 1 on 410,168 IPIP-NEO-120 respondents. S(const) is the best constant predictor; S(faithful) is a simulator that reproduces the cluster response distribution exactly. The faithful simulator is worse in every cluster.

ordering that §8 shows is inverted under the audited metric is *correct* under the proper one. Humans overtake the constant at $\lambda^\star$=0.30, comfortably below the proper coefficient. The model needs $\lambda^\star$=0.75 — a rule that over-rewards dispersion — and in 17 of 20 cells never overtakes the constant at any λ. So the simulator's failure is not an artefact of a bad metric: it survives the correct one.

Table 2 evaluates the bound on the real data: the structural penalty against a faithful simulator averages **7.75 points**, and $\mathrm{GMD}_j \geq \mathrm{MAD}_j$ holds on 120/120 items in all four clusters.

4 Empirical audit

Setup. We re-score a published cluster-level digital-twin system on its own artefacts: the same 410,168-respondent IPIP-NEO-120 corpus, the same k=4 clusters, and the exact train/test respondent IDs it recorded, for five LLMs (GigaChat3-10B, GPT-4.1-mini, GPT-4.1-nano, Grok-4.1-fast, Qwen3-235B) × four clusters. We add no LLM calls: model answers are read from the released run artefacts. Baselines are fitted on the recorded train IDs and scored on the recorded test IDs.

Baselines. *Cluster mean*: the per-item rounded mean of the cluster's training respondents, one constant vector. *Cluster majority*: the per-item mode. *Global mean*: the per-item grand mean over all respondents, ignoring clusters. *Empirical sample*: sample each item independently from the cluster's training response distribution. *Real human*: copy the answer vector of a randomly chosen training respondent from the same cluster.

Result. Table 3 reports S. The evolutionary persona averages 0.660 as released and 0.739 after the orientation correction of §7; the constant cluster mean averages 0.796. The LLM loses in 20 of 20 cells under either orientation, by 13.6 and 5.8 points respectively. The best single cell in the entire system (0.737) is below the *worst* cluster-mean baseline (0.779). The global mean — which never looks at the cluster — scores 0.720 and also beats every model cell.

The effect is below the protocol's noise floor. Re-drawing which 100 respondents are sampled, holding the 60/40 split ratio fixed, moves the *parameter-free* cluster-mean baseline by 1.3–2.5 points across 20 random draws. The improvement claimed for evolutionary optimisation in prior work is +0.72 points. With 40 test respondents per cell and a single fixed split, the reported effect is not separable from which rows landed in the test set.

5 What is a persona worth, in respondents?

"It loses to a constant" invites the reply that the constant is a strawman. So we restate the finding in a unit that is harder to dismiss: how many real respondents you would have to sample before a plug-in estimate of the cluster's response distribution matches what the persona gives you.

Drawing m real respondents, forming the per-item empirical distribution, and sampling a synthetic population from it yields distributional fidelity 0.723 at m=1, 0.802 at m=2, 0.933 at m=21 and 0.945 at m=34. Reading the audited systems off that curve:

System	DF	Resp.-equiv.
Evolutionary LLM persona	0.703	**1**
Constant cluster mean	0.796	2
Human ceiling	0.939	34

An evolutionary-optimised persona is worth about **one** real respondent: sampling a single human from the cluster gives a better estimate of that cluster's response distribution than the simulator does. This is the same result as §4 in different

Model	Cluster				Mean
	0	1	2	3	
GPT-4.1-mini	0.657	0.638	0.710	0.652	0.664
GPT-4.1-nano	0.673	0.678	0.697	0.670	0.679
GigaChat3-10B	0.711	0.691	0.737	0.672	0.703
Grok-4.1-fast	0.598	0.583	0.659	0.614	0.613
Qwen3-235B	0.634	0.615	0.677	0.642	0.642
Cluster mean (constant)	**0.794**	**0.827**	**0.785**	**0.779**	**0.796**
Real human, same cluster	0.720	0.747	0.715	0.704	0.721

Table 3: Answer similarity S on the original system's own test IDs. The constant cluster mean (bold) beats every LLM in every cell. A real human from the same cluster scores below the constant.

units, and it is the version we would put to a practitioner weighing synthetic respondents against a small pilot survey.

6 Does conditioning on the right cluster matter?

No persona study we know of reports the wrong-persona control, so no reported persona effect is currently attributable to the persona. We score every simulator against every cluster's humans — 80 cells — and test diagonal dominance by permutation.

Metric	Matched	Mismatched	Δ	p
S	0.652	0.631	$+0.021$	0.032
DF	0.698	0.672	$+0.026$	0.026
C2ST	0.982	0.981	-0.001	0.51
VR	0.431	0.436	-0.005	0.55

Conditioning on the correct cluster helps, slightly, on the two location-sensitive measures. It does nothing at all for distinguishability or dispersion: a persona scored against the *wrong* cluster's humans is as hard to tell from them as one scored against the right cluster's. This is the behavioural counterpart of §9 — the persona moves which items get endorsed, and not who is endorsing them — and it makes the wrong-persona control as necessary a baseline as the constant floor.

7 An orientation defect, and what survives it

Before comparing anything, the two sides must be on the same scale. They are not.

The released corpus stores human answers *reverse-recoded* so that a higher number always means more of the trait, while the simulator is shown the literal item text — "Cheat to get ahead" — and asked how accurately it describes it, i.e. it answers *raw*. On the 55 negatively-keyed items of the IPIP-NEO-120 the two scales run in opposite directions, and every published model-vs-human comparison is computed across that flip.

Two signatures make this unambiguous. Negatively-keyed items average 3.373 against 3.425 for positively-keyed ones — level, whereas raw self-reports on undesirable content sit far lower. And "Cheat to get ahead" carries a human mean of 4.15 out of 5.

The effect is large and it runs *against* the simulator. Correcting the orientation raises the correlation between the model's and the humans' per-item mean profiles from $+0.09$ — indistinguishable from no relationship at all — to $+0.63$, and raises S from 0.660 to 0.739.

What matters is which conclusions survive. Three do, and the invariances are stated before the numbers rather than after:

- **Proposition 1 is invariant.** MAD and GMD are computed on the human distribution alone, and both are reflection-invariant.
- **The variance ratio is exactly invariant**, since $\mathrm{sd}(6 - x) = \mathrm{sd}(x)$. The under-dispersion finding of §9 is untouched.
- **The constant still wins every cell.** Reorientation closes the gap from 13.6 to 5.8 points, but the constant cluster mean beats the corrected model in 20 of 20 cells.

Correcting the defect in fact *sharpens* the paper's claim. The corrected model scores 0.739, *above* the 0.721 of a real member of the target cluster — while a classifier separates that same model from real members at AUC 0.983 (§9). A metric that ranks a near-perfectly detectable simulator above the humans it fails to imitate is not measuring human-likeness under any reading.

We recommend an orientation check as a standing first step for benchmarks of this kind: correlate

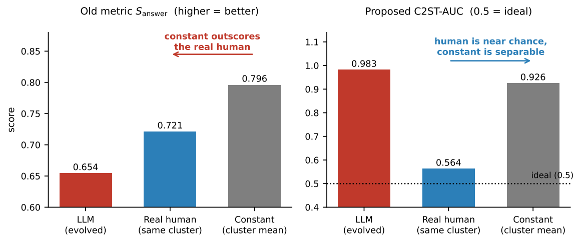

Figure 1: Left: under the old metric the constant outscores a real human. Right: under the proposed C2ST-AUC the human sits near chance (0.5, ideal) and the constant is near-perfectly separable. Only the right panel orders the three correctly.

the simulator's per-item mean profile against the population's, and treat a near-zero correlation as a scale defect until shown otherwise.

8 The human ceiling sits below the constant floor

The decisive check is what a *real* member of the target cluster scores. A cluster-level digital twin cannot do better than being an actual member of the cluster; that is the definition of success.

A randomly drawn same-cluster respondent scores 0.721 — and is beaten by the constant cluster mean in 20 of 20 cells (Table 3, Figure 1, left). This is exactly Proposition 1 realised: a real human *is* a draw from P_j, so their expected score is $1 - \mathrm{GMD}/4$, which the constant's $1 - \mathrm{MAD}/4$ must exceed. The predicted values (0.7996 constant, 0.7221 faithful) match the measured ones (0.796, 0.721) to within 0.4 points.

A measure that ranks a fixed vector above a genuine group member is not measuring group membership. Any ranking it induces over simulators — including the ranking prior work uses to select prompts — is therefore uninterpretable as evidence about personality fidelity.

9 A replacement suite

A usable metric must (i) be maximised by distributional fidelity rather than central tendency, and (ii) be reported against an explicit *human ceiling* and *constant floor*, so a number can be read as progress toward a real target rather than in the abstract. We propose three, computed from a set of simulated respondents rather than from paired individuals.

C2ST-AUC (primary). A classifier two-sample test: train a classifier to separate n simulated answer vectors from n held-out human vectors of the same cluster, and report cross-validated AUC. 0.5 means indistinguishable — the ideal twin. A constant is trivially separable. This directly operationalises the twin claim, and unlike S it cannot be gamed by collapsing onto the mean.

Distributional fidelity. $1 - \frac{1}{J}\sum_j W_1(\hat{P}_j^{\text{model}}, \hat{P}_j^{\text{human}})/4$, with W_1 the per-item Wasserstein-1 distance. Rewards matching the full response distribution.

Variance ratio. $\frac{1}{J}\sum_j \mathrm{sd}(\text{model}_j)/\mathrm{sd}(\text{human}_j)$; 1.0 is ideal, a constant gives 0. A diagnostic, not an optimisation target.

Results. Table 4 applies the suite to the same 20 cells. All three order human $>$ constant, as required; the old metric does the reverse. Two substantive findings follow. First, current personas are *near-perfectly separable* from real humans (C2ST-AUC 0.983 against a human ceiling of 0.564) — and are in fact *more* distinguishable than a constant vector (0.926), because they miss both the location and the spread of the human distribution. Second, they are systematically under-dispersed: $\mathrm{VR} = 0.43$, i.e. LLM personas produce fewer than half the response variance of the humans they simulate (Figure 2). Evolutionary optimisation moves none of these appreciably.

Under-dispersion is the wrong summary. A single variance ratio hides which variance is missing. Decomposing each answer matrix as $Y_{pj} = \mu + a_p + b_j + (ab)_{pj} + e$ — respondent main effect, item main effect, and the person-by-item interaction that carries individuation — tells a sharper story. The simulator emits essentially the *right total* variance (ratio 1.005 over the 20 cells), but places it in the wrong component: the item main effect takes 81.7% of its variance against 29.8% for humans, while the interaction takes 16.5% against 67.0%. The interaction variance ratio is 0.458, and it is 0 for a constant by construction.

So the simulator is not emitting too little. It is emitting a confident cluster-level *answer key* — which items are endorsable — and almost none of which *person* endorses which item. We recommend reporting the interaction ratio rather than the plain variance ratio, since the two rank systems differently and only the former measures the quantity a personality simulator exists to reproduce.

Metric	LLM	Human ceiling	Const. floor
C2ST-AUC ↓	0.983	0.564	0.926
DF ↑	0.703	0.939	0.795
VR (1.0 ideal)	0.431	1.018	0.000
S_{answer} ↑ (old)	0.654	0.721	0.796 †

Table 4: The suite over 20 cluster × model cells, each metric against its human ceiling and constant floor. †: under the old metric the constant floor *exceeds* the human ceiling — the failure this paper documents.

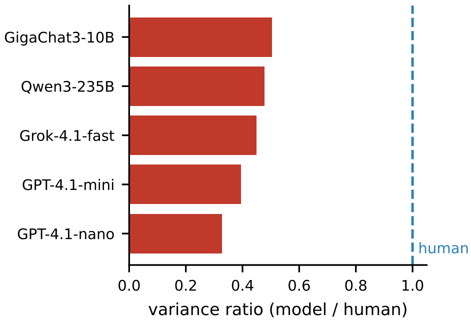

Figure 2: Response variance relative to the humans being simulated. Every model is substantially under-dispersed.

9.1 What is C2ST actually reading?

A metric that separates simulators from humans at AUC 0.983 invites an obvious objection: perhaps the classifier is detecting formatting rather than anything about personality. We test our own metric against it with a feature ladder, each rung scored identically and against the same anchors.

Features	LLM	Human	Const.
Full 120 items	0.983	0.564	0.926
5 style scalars only	0.950	0.566	0.994
Items, ipsatised	0.975	0.566	0.986
30 facet scores only	0.987	0.581	0.944
Forward-keyed only	0.990	0.568	0.810
Reverse-keyed only	0.984	0.583	0.843

Five content-free scalars — per-respondent mean and standard deviation, extreme -response rate, longest identical run, and forward-versus-reverse consistency — recover 93% of the full separability above chance. So the objection has real force, and we state it plainly: **most of what C2ST detects is response style.**

It is not only style. Ipsatising each respondent (removing their own mean and standard deviation, i.e. the two largest style features) still leaves AUC 0.975, and facet scores alone give 0.987. Style and content are redundant routes to the same conclusion rather than one masking the other. The human ceiling is also stable at 0.56–0.58 across every rung, so the anchor does not move with the feature set.

We therefore recommend reporting C2ST with its style-only rung alongside it, so readers can see how much of any separability is carried by response style. A simulator that closed only the style gap would look far better on the headline number while remaining just as distinguishable on content.

9.2 How much of the anchors is finite-sample bias?

Both anchors are estimated from n=40 per side, which for a classifier over 120 features is a small-sample regime. We sweep n on human-only data, where the true answers are known: two disjoint samples from the same cluster must be indistinguishable, so the honest C2ST value is exactly 0.5 and any excess is estimator bias.

n	20	40	80	160	320	640
C2ST human	—	0.574	0.542	0.536	0.527	0.517
DF human	0.915	0.939	0.957	0.969	0.978	0.985
C2ST const.	—	0.914	1.000	1.000	1.000	1.000

Two corrections follow, in opposite directions, and both land on their theoretical values. Fitting $y = y_\infty + cn^{-1/2}$ over the non-degenerate range gives an extrapolated C2ST human ceiling of **0.500** and an extrapolated DF human ceiling of **1.000** — exactly what two samples from the same distribution must score. So the C2ST ceiling of 0.574 at n=40 is *entirely* finite-sample classifier bias and should be read with its n attached, while the DF ceiling of 0.939 is *entirely* plug-in Wasserstein bias and is pessimistic. Correcting the latter widens the gap to the simulator's 0.703 rather than narrowing it.

Below $n \approx 40$ the suite is unusable: the classifier fails to separate even the constant from real humans (the dashed entries above are the degenerate 0.5), so it cannot be trusted to separate anything else. We therefore recommend $n \geq 40$ per side as a floor, and reporting every anchor together with the n it was measured at.

10 Is the prompt encoding the constraint?

A natural defence of the simulator is that the persona prompt simply cannot carry enough information. In this pipeline an individual reaches the model as a quantised code: each of 5 traits and 30 facets contributes one adverb drawn from a five-level scale, attached to text that is otherwise constant within a cluster. The whole individual is 35 symbols over a five-letter alphabet.

We test the defence by asking what an *oracle* reader of that code could do. Fitting on half of each cluster and scoring on the other half, two oracles differ only in what they predict: a **mean** oracle (ridge on the ordinal code), which predicts the conditional mean; and a **faithful** oracle, the same fit plus a resampled residual, which reproduces the conditional *distribution*.

The faithful oracle attains a variance ratio of 0.988 — essentially human. So the encoding is not the binding constraint: 35 symbols suffice for human-level dispersion, and the shortfall in §11 belongs to the model, not to the prompt. Codes are also effectively unique (collision rate 0), so quantisation is not merging individuals either.

The two oracles also reproduce Proposition 1 inside a single experiment. The mean oracle scores $S = 0.802$; the faithful one, which is strictly closer to the truth, scores $S = 0.730$ — worse by 7.2 points, against the 7.75 the GMD − MAD gap predicts. Predicting the distribution correctly is penalised, and by the predicted amount.

One further number bounds what conditioning buys at all: the mean oracle beats the constant (0.792) by 0.9 points. A perfect reader of the persona gains under a point over ignoring it.

11 Where the variance is lost

Knowing the simulator is mis-dispersed does not say *where* the dispersion goes. We read the model's own belief before any sampling occurs, on open-weight models served locally.

Stage 1, belief. For each persona and item we take the model's next-token distribution over the answer tokens $\{1, \ldots, 5\}$ directly from the logprobs. Two implementation details matter. Hybrid reasoning models emit a <think> token first — with probability ≈ 1 — so a naive readout sees no mass on the scale at all; we disable thinking and prefill the assistant turn.

Stage 2, decode. We then sample each item from its own belief distribution and score the resulting population.

Result. The belief distribution is already collapsed.

The statistic must be chosen carefully. The quantity comparable to a human *across-respondent* SD is not the average within-persona conditional SD, but the SD of the persona-*mixture* distribution, obtained from the law of total variance,

$$\mathrm{Var}(X_j) = \mathbb{E}_i[\mathrm{Var}(X_j \mid i)] + \mathrm{Var}_i(\mathbb{E}[X_j \mid i]) \, ,$$

computed exactly from the retained probability vectors. Reporting the within-persona term alone would understate a faithful simulator, which can hold sharp per-persona beliefs and still match human variance through differing persona means.

Model	Params	$\mathrm{VR}_{\mathrm{with}}$	$\mathrm{VR}_{\mathrm{betw}}$	$\mathrm{VR}_{\mathrm{tot}}$
Qwen2.5-7B-Instruct	7B	0.367	**0.193**	0.430
Mistral-Small-24B	24B	0.538	**0.177**	0.579
Qwen3-235B-A22B-2507	235B	0.695	**0.258**	0.758
human	—	—	—	1.000

Essentially all probability mass lands on the five valid answer tokens (0.97–1.00), so no model has difficulty *emitting* a number; entropies of 0.42–0.73 nats sit far below the maximum $\ln 5 = 1.609$. The largest model is the one the audited system itself used, so this is a measurement on that system, not on a proxy for it.

Dispersion scales; individuation barely does. The decomposition says something the total does not. Across a $30\times$ range in parameters and two model generations, total dispersion rises substantially, $0.430 \rightarrow 0.579 \rightarrow 0.758$. But that rise is carried almost entirely by the *within*-persona term — per-item uncertainty — which nearly doubles, $0.367 \rightarrow 0.695$. The *between*-persona term, the part that actually individuates, stays in a narrow band: 0.193, 0.177, 0.258, never above about a quarter of the human value, and not monotone in scale.

A simulator therefore looks markedly more dispersed largely by being less confident, not by personalising better. This has a direct reporting consequence: $\mathrm{VR}_{\mathrm{total}}$ confounds uncertainty with individuation, and $\mathrm{VR}_{\mathrm{between}}$ is the quantity a persona claim should be judged on.

We state the limit of this plainly. At $\mathrm{VR}_{\mathrm{total}} = 0.758$ the largest model is no longer far from the human marginal, so for it the strong form of the claim below — that no sampling-side remedy could help — is not supported; what remains unsupported at every scale we measured is *individuation*.

Three consequences follow. First, for the smaller models the dispersion is not discarded at the decoding step; it was never present, so no temperature, top-p, or sampling-side remedy could recover it. Second, because all probability mass lands on valid answer tokens, accounts attributing mode collapse to difficulty *emitting* numeric scores are not supported here — the model is not failing to express a broad belief, it holds a narrow one. Third, this closes the loop with the answer-level results: the item main effect dominates realised variance (§9), conditioning on the correct cluster does nothing for dispersion or distinguishability (§6), and the belief-level individuating component is ~ 0.18 of human in both models. Three independent measurements agree that persona conditioning moves *which items* get endorsed, not *who* is endorsing them.

12 Is $k{=}4$ a partition at all?

The systems we audit condition on a $k{=}4$ k-means partition of the 30 facet scores. We re-derive it and test $k \in \{2, \ldots, 15\}$ under three preprocessings. Our $k{=}4$ solution reproduces the published labels (ARI $= 0.946$), confirming the pipeline.

No *separation* criterion supports $k{=}4$. Silhouette falls monotonically from 0.134 at $k{=}2$ to 0.053 at $k{=}15$, taking 0.082 at $k{=}4$; Davies–Bouldin is also minimised at $k{=}2$; the gap statistic increases monotonically and selects no interior k. A silhouette below 0.1 indicates essentially no separated structure.

Stability tells a different and fairer story, and we report it because it partly vindicates the choice. Bootstrap adjusted Rand index over 40 resamples is 0.977, 0.952, 0.932 and 0.903 at $k{=}2, 3, 4, 5$, then falls away sharply (0.797 at $k{=}6$, 0.581 at $k{=}7$, and below 0.7 everywhere after). So $k{=}4$ is close to the largest k that k-means recovers reproducibly, even though it never separates. The clusters are also not demographic artefacts: Cramér's V against sex, age band and country is 0.055–0.057, detectable at $n{=}410{,}168$ but negligible in size.

The resolution is that stability and separation measure different things. A reproducible partition of a continuum is still a partition of a continuum, and that is what bounds how much cluster-conditional signal any persona can exploit — as §6 then confirms behaviourally.

13 Related work

Personality conditioning of LLMs is commonly validated by administering psychometric inventories to the conditioned model (Serapio-García et al., 2023; Jiang et al., 2024), and prompt sensitivity of the resulting traits is well documented (Dong et al., 2025; Gupta et al., 2024). Individual-level simulation from interview transcripts achieves high behavioural fidelity (Park et al., 2024) at a per-person cost that motivates the cluster-level shortcut we study. Evolutionary and search-based prompt optimisation (Guo et al., 2024; Prasad et al., 2023; Zhou et al., 2022) supplies the optimiser. Our contribution is orthogonal to all of these: we show the objective they are scored against is maximised by a degenerate solution.

Classifier two-sample tests are standard for comparing distributions (Lopez-Paz and Oquab, 2017), and the failure mode we identify — an absolute-error objective minimised by a conditional central tendency — is the familiar reason such losses produce blurred, low-variance predictions in generative modelling.

14 Conclusion

Cluster-level LLM personality simulation is evaluated with a metric that a constant vector wins and a real human loses. We proved why, measured the structural penalty at 7.75 points, confirmed the inversion in 20 of 20 cells across five models, and proposed a suite that orders humans above constants. Reported gains in this literature that are smaller than the constant-vs-faithful gap should not be read as evidence of personality fidelity. The repair is equally narrow: score with $S_{1/2}$, the proper member of the family in (2), and report the constant floor and the human ceiling next to every number.

Reproducibility

Every analysis in this paper is deterministic given the released corpus and the committed run artefacts, and requires no LLM calls except the belief-level measurement of §11. We ran the full analysis pipeline independently on two clusters with different operating systems, Python versions and library versions (scikit-learn 1.2 vs 1.7, pandas 2.1 vs 3.0). Every reported figure — the $\lambda^{\star}$ values, the orientation deltas, the variance components, and the oracle ceilings — agreed to all printed digits. Code and per-cell results are released.

Limitations

Our audit covers one corpus (IPIP-NEO-120), one clustering pipeline, one optimisation family, and five models; the belief-level measurement of §11 covers three open-weight models spanning 7B–235B and two generations, since it requires log-prob access the API models in the audit do not provide; the scaling statement rests on three points and should be read as a direction, not a fitted law; the proposition is general to the absolute-error metric in (1), but the empirical magnitudes are specific to this setting. The model answers we re-score were produced by prior runs we did not control, at temperatures we did not set, and one released run was committed with entirely missing answers and is excluded. Our human ceiling is a single same-cluster respondent, which underestimates what a well-specified individual-level twin could achieve; we make no claim about individual-level methods. C2ST-AUC depends on classifier capacity and on n=40 per side, so its absolute value should be read as a lower bound on separability, and §9.1 shows most of what it reads is response style; and DF treats items independently and so ignores within-respondent answer dependence. The corpus is self-reported, skewed toward respondents aged 18–25 and toward the USA, UK and Canada, so neither the clusters nor the ceilings should be read as representative of any broader population. Finally, we show current personas fail this evaluation; we do not show that any method can pass it.

Ethics statement

We use a public, de-identified corpus of self-reported personality questionnaire responses and add no new human-subjects data collection. Our results are cautionary: they indicate that current cluster-level personality simulations are readily distinguishable from real respondents and under-represent human response variance, and so should not be used as substitutes for human participants in research, product, or policy decisions. Reporting an evaluation number without its constant floor risks overstating the fidelity of synthetic respondents, which is the specific harm this paper aims to reduce. Simulating demographic or personality groups also risks reinforcing stereotypes when cluster descriptions are treated as characterisations of real people rather than as statistical summaries.

References

Yifan Dong and 1 others. 2025. Can large language model agents reliably express personality? *arXiv preprint*.

Qingyan Guo, Rui Wang, Junliang Guo, Bei Li, Kaitao Song, Xu Tan, Guoqing Liu, Jiang Bian, and Yujiu Yang. 2024. Connecting large language models with evolutionary algorithms yields powerful prompt optimizers. In *International Conference on Learning Representations*.

Shashank Gupta, Vaishnavi Shrivastava, Ameet Deshpande, Ashwin Kalyan, Peter Clark, Ashish Sabharwal, and Tushar Khot. 2024. Bias runs deep: Implicit reasoning biases in persona-assigned language models. In *International Conference on Learning Representations*.

Guangyuan Jiang, Manjie Xu, Song-Chun Zhu, Wenjuan Han, Chi Zhang, and Yixin Zhu. 2024. Evaluating and inducing personality in pre-trained language models. In *Advances in Neural Information Processing Systems*.

John A. Johnson. 2014. Measuring thirty facets of the five factor model with a 120-item public domain inventory: Development of the ipip-neo-120. *Journal of Research in Personality*, 51:78–89.

David Lopez-Paz and Maxime Oquab. 2017. Revisiting classifier two-sample tests. In *International Conference on Learning Representations*.

Joon Sung Park, Carolyn Q. Zou, Aaron Shaw, Benjamin Mako Hill, Carrie Cai, Meredith Ringel Morris, Robb Willer, Percy Liang, and Michael S. Bernstein. 2024. Generative agent simulations of 1,000 people. In *arXiv preprint arXiv:2411.10109*.

Archiki Prasad, Peter Hase, Xiang Zhou, and Mohit Bansal. 2023. GrIPS: Gradient-free, edit-based instruction search for prompting large language models. In *Proceedings of the 17th Conference of the European Chapter of the ACL*.

Greg Serapio-García, Mustafa Safdari, Clément Crepy, Luning Sun, Stephen Fitz, Peter Romero, Marwa Abdulhai, Aleksandra Faust, and Maja Matarić. 2023. Personality traits in large language models. *arXiv preprint arXiv:2307.00184*.

Yongchao Zhou, Andrei Ioan Muresanu, Ziwen Han, Keiran Paster, Silviu Pitis, Harris Chan, and Jimmy Ba. 2022. Large language models are human-level prompt engineers. In *International Conference on Learning Representations*.